



BOHLOKO FAMILY FARM

POULTRY PROCESSING & SALES PLATFORM

From Farm to Fork — A Complete Digital poultry Management Ecosystem

South Africa's poultry industry faces critical operational challenges

Small-to-medium poultry farms lose revenue through inefficiency, waste, and lack of visibility.

No Real-Time Visibility

Farmers lack real-time data on inventory levels, production cycles, and sales — leading to stockouts, overstocking, and spoiled perishable goods.

Manual & Paper-Based

Critical operations — feed tracking, medication logs, quality checks — are still done on paper, causing errors, lost data, and compliance risks.

Revenue Leakage

Inefficient order processing, no customer loyalty programs, and poor sales channel management result in lost repeat business and revenue.

Compliance Burden

Food safety regulations (DAFF, POPIA) require documented audit trails — manual systems can't keep up, risking fines and shutdowns.

Growth Barriers

Scaling from 500 to 5,000+ birds per cycle requires systems that manual processes simply cannot support.

Disconnected Systems

No integration between production, inventory, sales, and finance — creating data silos and decision-making delays.

An end-to-end digital platform for poultry production, processing, and sales

Production Management

- Plan & track broiler production cycles
- Daily logging: feed, water, mortality
- Medication & vaccination tracking
- Automated mortality alerts (>5%)
- Feed conversion ratio analytics

Inventory & Processing

- Harvest into 10 cut types (whole, breast, wings, etc.)
- Real-time stock level monitoring
- Expiry tracking with automated alerts
- Batch traceability: farm to sale
- Multi-location storage management

Sales & Orders

- Multi-channel ordering (farm gate, online, wholesale)
- 6 payment methods including mobile money
- Automated receipts & invoicing
- Order status tracking & notifications
- Cash-on-delivery verification

Customer Relationship

- 5-tier loyalty program (Bronze → Diamond)
- Customer segmentation & targeting
- Promotional campaigns (email, SMS, push)
- Feedback & complaint management
- Customer lifetime value analytics

Quality & Compliance

- 9 quality check types (microbiological, chemical, etc.)
- Corrective action tracking
- Quality certificate generation
- Full audit trail for regulatory inspections
- POPIA & food safety compliance

Analytics & Reporting

- Production performance dashboards
- Sales & financial analytics
- Inventory turnover & aging reports
- Export to PDF, Excel, CSV
- Forecasting & trend analysis

R 25B+

SA Poultry Market Value

~60%

of SA Chicken Consumption

2,500+

Small Poultry Farms in SA

<15%

Farms Using Digital Systems

Target Customer Segments

- ✓ **Consumers** — Direct farm-to-table chicken buyers
- ✓ **Restaurants** — Bulk fresh chicken orders with traceability
- ✓ **Retailers** — Supermarkets and butcheries needing reliable supply
- ✓ **Distributors** — Wholesale buyers requiring volume and logistics
- ✓ **Farm Gate Buyers** — Walk-in customers at the farm
- ✓ **Institutions** — Schools, hospitals, and caterers

Market Drivers

- ✓ Growing demand for traceable, locally-sourced poultry
- ✓ Increasing regulatory requirements for food safety
- ✓ Mobile money adoption accelerating in rural SA
- ✓ Rising consumer preference for farm-fresh products
- ✓ Government support for agricultural digitisation
- ✓ Need to reduce R3.5B annual poultry import bill

KEY PARTNERS		KEY ACTIVITIES		VALUE PROPOSITIONS		CUSTOMER RELATIONSHIPS		CUSTOMER SEGMENTS	
• Firebase / Google Cloud • Stripe, PayFast, Yoco • DAFF (Dept. of Agriculture) • Feed suppliers • Cold chain logistics partners • Local municipalities		• Poultry production management • Processing & packaging • Quality control & compliance • Order fulfilment & delivery • Customer relationship management • Data analytics & reporting		• Complete farm-to-fork traceability • Real-time inventory visibility • Automated compliance documentation • Multi-channel sales platform • Food safety & quality assurance • Data-driven decision making		• Self-service online ordering • 5-tier loyalty program • Promotional campaigns • Feedback & complaint tracking • Personalised recommendations • Dedicated account management (B2B)		• Consumers (direct buyers) • Restaurants & caterers • Retailers & butcheries • Wholesale distributors • Farm gate customers • Institutions (schools, hospitals)	
KEY RESOURCES			CHANNELS			REVENUE STREAMS			
• Poultry farm infrastructure • Processing facility • Digital platform (web app) • Skilled workforce • Customer data & analytics			• Farm gate (walk-in sales) • Online ordering platform • Wholesale distribution network • Mobile app (planned) • Social media & digital marketing • SMS & email promotions			• Product sales: Whole chicken, portions, value-added cuts • B2B contracts: Restaurant & retailer supply agreements • Wholesale volume: Bulk orders with tiered pricing • Value-added products: Pre-packed, marinated, seasoned cuts • Future: Subscription boxes, export sales			
COST STRUCTURE						COST HIGHLIGHTS			
• Feed & medication (largest variable cost ~60%) • Labor (processing, sales, farm staff) • Platform hosting & maintenance (R5,000/month) • Cold storage & logistics • Marketing & customer acquisition • Regulatory compliance & certification						• Initial Investment: R150,000 • Monthly Operating: R5,000 • Payback Period: 18 months • Feed costs dominate: ~60% of production cost			

37

Frontend Pages

19

API Routes

15

Data Models

9

Modules Fully Implemented

Production Module

- ✓ Cycle planning & approval workflow
- ✓ Daily production logs
- ✓ Medication & vaccination tracking
- ✓ Mortality rate monitoring
- ✓ Harvest processing (10 cut types)

Inventory Module

- ✓ Real-time stock monitoring
- ✓ Multi-location management
- ✓ Expiry date tracking & alerts
- ✓ Batch traceability
- ✓ Transfer workflow management

Orders & Payments

- ✓ Product catalogue with real-time stock
- ✓ 6 payment methods
- ✓ 3 payment gateway integrations
- ✓ Partial payments & payment plans
- ✓ Automated receipts & invoices

CRM Module

- ✓ Customer profiles & order history
- ✓ 5-tier loyalty program
- ✓ Promotional campaign management
- ✓ Customer segmentation
- ✓ Customer lifetime value analytics

Quality Control

- ✓ 9 quality check types
- ✓ Corrective action tracking
- ✓ Certificate generation
- ✓ Full audit trails
- ✓ Scheduled quality checks

Analytics & Admin

- ✓ Production & sales dashboards
- ✓ Inventory turnover reports
- ✓ System configuration panel
- ✓ User role management (RBAC)
- ✓ Data backup & export tools

Backend Architecture

Node.js TypeScript Express.js Firebase Auth Firestore JWT
bcrypt

Microservices-ready architecture with clean separation of concerns. Service layer pattern with repository abstraction and dependency injection.

Frontend

React TypeScript Material-UI Axios Responsive

37-page responsive dashboard with role-based views. Farm-themed UI with offline capability for field staff.

Infrastructure

Google Cloud Firebase Docker CI/CD

Cloud-native deployment with auto-scaling, daily backups, and disaster recovery (4-hour RTO).

Design Principles

- ✓ **SOLID Principles** — SRP, OCP, LSP, ISP, DIP enforced across all modules
- ✓ **Service Layer Pattern** — Business logic separated from data access
- ✓ **Repository Pattern** — Data access abstracted for testability
- ✓ **DTO Pattern** — Clean API contracts between frontend and backend
- ✓ **Dependency Injection** — Loose coupling for maintainability
- ✓ **Strategy Pattern** — Pluggable payment gateways
- ✓ **Observer Pattern** — Event-driven notifications

Performance Targets

- Page load: ≤ 3 seconds
- API response: ≤ 1 second (95th percentile)
- Uptime: 99.5% during business hours
- 50→200 concurrent users (scalable)

What Sets Us Apart

- ✓ **End-to-end integration** — Only platform covering production → processing → sales in one system
- ✓ **South African first** — Built for SA regulations, payment methods (PayFast, Yoco, mobile money), and currency (ZAR)
- ✓ **Offline-capable** — Works in rural areas with limited connectivity
- ✓ **Bilingual** — English & Sesotho language support
- ✓ **Mobile-first** — Designed for field staff with smartphones

Compliance Built-In

- ✓ POPIA-compliant data handling
- ✓ Food safety audit trails (DAFF)
- ✓ Automated quality certificates
- ✓ 7-year financial record retention
- ✓ VAT calculation & reporting

Cost Efficiency

R150K

Initial Investment

R5K

Monthly Operating

18 mo

Payback Period

Reduces manual data entry by **60%** and improves inventory accuracy to **99%**. ROI driven by efficiency gains, waste reduction, and increased sales.

Measurable Impact

- 60% reduction in manual data entry
- 99% inventory accuracy (up from ~80%)
- Real-time visibility across all operations
- Faster order processing (<30 seconds)
- Automated compliance reporting
- Scalable from 500 to 5,000+ birds/cycle

R150,000

Total Development Cost

R5,000

Monthly Operating Cost

R240,000+

Year 1 Revenue Target

18 Months

Investment Payback

Revenue Model

- ✓ **Direct Sales** — Farm gate, online orders for whole & portioned chicken
- ✓ **B2B Contracts** — Recurring supply agreements with restaurants & retailers
- ✓ **Wholesale Volume** — Tiered pricing for bulk buyers (distributors, institutions)
- ✓ **Value-Added Products** — Pre-packed, marinated, and seasoned cuts at premium pricing
- ✓ **Future: Subscriptions** — Weekly/monthly chicken box subscriptions for consumers

Cost Breakdown

- ✓ **Feed & Medication:** ~60% of production costs (largest variable cost)
- ✓ **Labor:** Processing staff, farm attendants, sales team
- ✓ **Platform Hosting:** R5,000/month (Firebase/Google Cloud)
- ✓ **Cold Storage:** Refrigeration and freezer operations
- ✓ **Marketing:** Digital marketing and promotional campaigns
- ✓ **Compliance:** Certifications, inspections, regulatory fees

Key Metric: System must pay for itself within 18 months through efficiency gains, reduced waste, and increased sales volume.

Phase 1 — Months 1-3

Core Production & Inventory

- Production cycle management
- Daily logging & medication tracking
- Harvest processing (10 cut types)
- Real-time inventory tracking
- Storage location management

Phase 2 — Months 4-5

Orders & Sales

- Product catalogue
- Multi-channel ordering
- Payment processing (6 methods)
- Receipt & invoice generation
- Order tracking & notifications

Phase 3 — Months 6-7

Analytics & CRM

- Production analytics dashboards
- Sales & financial reports
- CRM system (loyalty, campaigns)
- Customer segmentation
- Inventory analytics

Phase 4 — Months 8-9

Compliance & Mobile

- Quality control system
- Compliance reporting
- Certificate generation
- Mobile app for field staff
- Offline capability

9-Month Total Timeline

Phased delivery ensures core operations are live within 3 months, with full platform completion by month 9. Agile methodology with regular stakeholder demos.

Future Enhancements

AI-powered recommendations, predictive analytics, supplier management, barcode/QR scanning, accounting system integration, and mobile money expansion.



Thank You

LET'S BUILD THE FUTURE OF POULTRY FARMING TOGETHER

Investment Required

R 150,000

Payback Period

18 Months

Market Readiness

90% Complete

Bohloko Family Farm — Digitising South African Poultry, One Farm at a Time